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“Rebuilding” Statistics in the Age of AI: Culture, Infrastructure, and Training

David L. Donoho^a, Jian Kang^b , Xihong Lin^{c,d,e}, Bhramar Mukherjee^{f,g}, Dan Nettleton^h, Rebecca Nugentⁱ , Abel Rodriguez^j, Eric P. Xing^{k,l}, Tian Zheng^m, and Hongtu Zhuⁿ

^aDepartment of Statistics, Stanford University, Stanford, CA; ^bDepartment of Biostatistics, University of Michigan, Ann Arbor, MI; ^cHarvard T.H. Chan School of Public Health, Boston, MA; ^dDepartment of Statistics, Harvard University, Cambridge, MA; ^eBroad Institute, Cambridge, MA; ^fYale School of Public Health, New Haven, CT; ^gDepartment of Statistics and Data Science, Yale University, New Haven, CT; ^hDepartment of Statistics, Iowa State University, Ames, IA; ⁱDepartment of Statistics and Data Science, Carnegie Mellon University, Pittsburgh, PA; ^jBaskin School of Engineering, University of California, Santa Cruz, Santa Cruz, CA; ^kMohamed bin Zayed University of Artificial Intelligence, Abu Dhabi, United Arab Emirates; ^lSchool of Computer Science, Carnegie Mellon University, Pittsburgh, PA; ^mDepartment of Statistics, Columbia University, New York, NY; ⁿDepartment of Biostatistics, University of North Carolina at Chapel Hill, Chapel Hill, NC

ABSTRACT

This perspective article distills themes from the 2024 JSM town hall, Statistics in the Age of AI, where panelists discussed how the field should evolve as foundation models, large-scale empirical modeling, and data-intensive infrastructures reshape science and society. Rather than a transcript or comprehensive survey, the article offers a forward-looking perspective grounded in the town hall discussion and the authors’ reflections. We organize the discussion around five recurring questions: (i) disciplinary culture and incentives, (ii) the role of data curation and annotation, (iii) engagement with modern empirical modeling, (iv) training for large-scale AI applications, and (v) partnerships with key AI stakeholders. We emphasize that statistics already leads in areas such as uncertainty quantification, study design, evaluation, and inference, while arguing that some parts of the field could engage more deeply with data engineering, cloud-scale computation, and end-to-end AI systems. We conclude with pragmatic recommendations for training, incentives, and collaboration so that statisticians can help shape the data-centric future.

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1. Introduction

The rapid expansion of artificial intelligence (AI) and large-scale data technologies is transforming how scientific research, industry, and society operate (Goodfellow et al. 2016; Vaswani et al. 2017; Moor et al. 2023; Pearson et al. 2025; US.NSF.Inf 2025; US.NIH.AI 2025; US.NSF.NAIRI 2025; The White House 2025; National Academies of Sciences, Engineering, and Medicine 2026). Foundation models, large-scale empirical modeling, and data-intensive infrastructures now shape decision-making in domains ranging from health and the environment to economics and public policy. These developments bring both opportunities and pressures for the statistical community (Breiman 2001; He et al. 2025; Lin et al. 2025; Cai 2025; American Statistical Association 2025): statistics remains central to uncertainty quantification, causal reasoning, and principled evaluation (National Academies of Sciences, Engineering, and Medicine 2026), yet many of our traditional practices, institutional structures, and training models were not designed for the current scale, speed, and engineering demands of AI.

To reflect on these shifts and to stimulate a broader community conversation, we organized a town hall session, *Statistics in the Age of AI: a Town Hall with an Expert Panel*, in the 2024 Joint Statistical Meetings (JSM). The session brought

together a distinguished panel of statisticians, including Drs. David Donoho, Xihong Lin, Bhramar Mukherjee, Dan Nettleton, Rebecca Nugent, Abel Rodriguez, and Eric Xing, and was moderated by Drs. Tian Zheng and Hongtu Zhu. Rather than formal talks, the town hall relied on open discussion and audience Q&A, with the aim of surfacing candid perspectives on how statistics, statistics/biostatistics departments, and statisticians are being empowered, challenged, and reshaped by AI.

Figure 1 provides a visual summary of several cross-cutting themes that emerged repeatedly during the town hall discussion. These include a shift from narrowly scoped methodological contributions toward end-to-end, system-level problem solving; explicit recognition of data curation, annotation, and engineering as foundational components of trustworthy AI; deeper engagement with modern empirical and algorithmic modeling; modernization of training around communication, collaboration, and computation; and clearer articulation of statistics’ distinctive strengths, such as uncertainty quantification, bias assessment, causal reasoning, and rigorous evaluation, to a broad range of stakeholders.

This perspective article grows out of that town hall and is intended as a forward-looking reflection rather than a transcript or a comprehensive survey. Our goal is not to offer an exhaustive account of the many ways statistics already engages with AI, nor

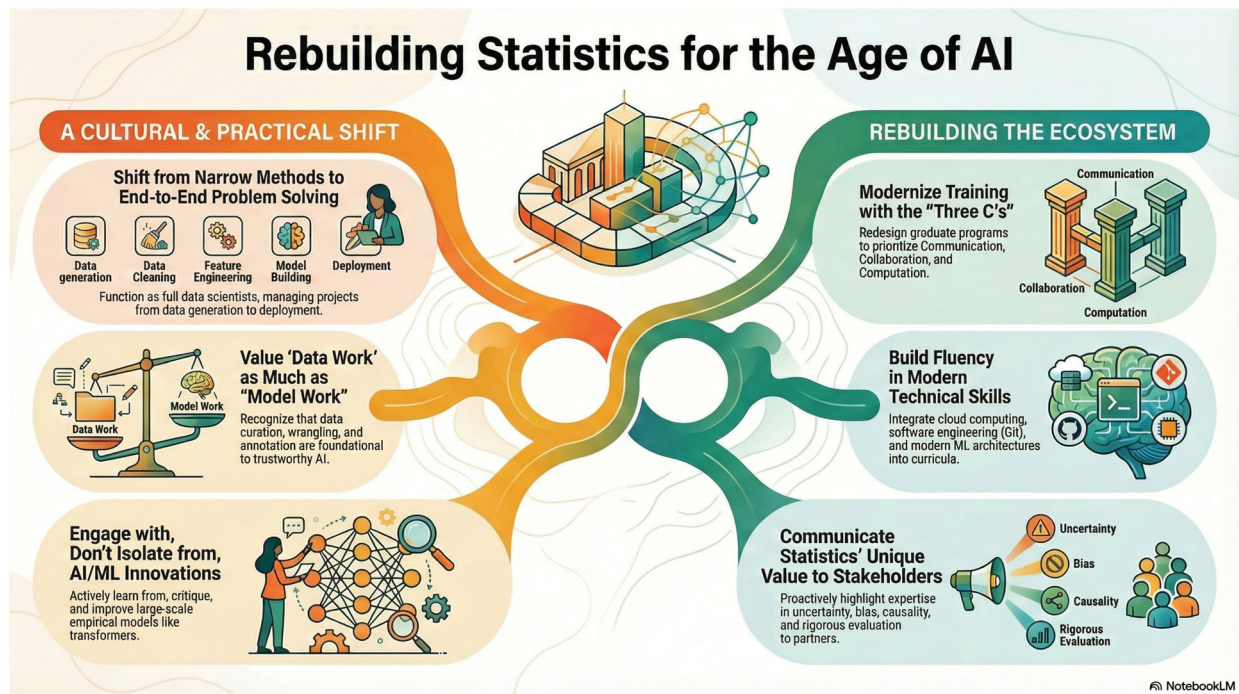


Figure 1. Key themes emerging from the 2024 JSM town hall on “Statistics in the Age of AI,” including system-level problem solving, the central role of data work, engagement with modern empirical modeling, and the need to modernize training, collaboration, and institutional positioning.

to make uniform claims about the field as a whole. Instead, we synthesize recurring themes from the discussion and highlight areas where, in our view, statistics has important opportunities to build on existing strengths, address uneven progress across subfields, and expand its role in the age of AI. We organize the article around five interrelated questions that repeatedly emerged during the session:

- *Question (i) (Culture and Practices of Statistics):* How can we optimize our culture, practices, and infrastructure to better engage with emerging large-scale data processing and analysis needs? What primary challenges must statistics address to stay relevant and impactful?
- *Question (ii) (Data Curation and Data Work):* How can statisticians ensure a deeper understanding of data, including data curation, wrangling, and annotation, and recognize the central role of this data work” in method development and AI systems?
- *Question (iii) (Engaging with Innovations in Empirical Modeling):* Can the statistical community be actively engaged in understanding, critiquing, and improving recent innovations in machine learning and empirical modeling, rather than treating them as external or purely engineering developments?
- *Question (iv) (Next-Generation Talent in Large-Scale Applications):* What are the best practices for training the next generation of statisticians for research in large-scale data and AI applications? What successful programs or initiatives have helped nurture the necessary skills and expertise?
- *Question (v) (Engaging AI Stakeholders):* How can we promote statistics and its values to AI stakeholders, including academia, industry, funding agencies, and policymakers, while fostering talent, securing resources, and building a robust ecosystem and collaboration network?

The remainder of the article is organized around these five questions. Each subsequent section takes up one of these themes, synthesizing insights from the town hall discussion and extending them through our own reflections on their implications for research, training, and institutional development. For clarity, these sections present concise versions of the panelists’ original remarks and conversation. The full town hall conversation is available in a companion arXiv preprint. Our goal is to encourage a forward-looking, pragmatic, and system-level conversation about how statisticians can reshape their culture, tools, training practices, and institutional roles to lead in the data- and AI-centric future.

2. Culture and Practices of Statistics

The culture and everyday practices of statistics must evolve to remain impactful in the age of AI. Rather than treating AI as an external threat or a separate discipline, the panel emphasized that statisticians should rethink their roles, workflows, and technical repertoire within today’s large-scale, engineering-heavy data ecosystems (US.NSF.Inf 2025; US.NIH.AI 2025; US.NSF.NAIRI 2025; The White House 2025).

A useful starting point is to frame statistics, first and foremost, as the science of learning from data to enable real-world prediction and discovery (Breiman 2001; He et al. 2025; Lin et al. 2025; Cai 2025). That mission spans both small, carefully designed studies and modern large-scale datasets. Accordingly, our foundational strengths, including experimental design, mixed-effects modeling, and other “laptop-scale” tools, remain indispensable for everyday scientific practice. What most clearly distinguishes statisticians, however, is not allegiance to any single model class but a disciplined approach to uncertainty management: moving beyond point prediction

to interval estimation, principled evaluation, and transparent statements about what the data can and cannot support. Seen through this lens, the reproducibility crisis is fundamentally an uncertainty-management crisis, and attention to data collection, bias, missingness, confounding/causality, and dependence structure becomes central rather than peripheral (Markowetz 2024; Van Calster et al. 2026).

It is important to emphasize that these patterns are not uniform across the field. Many areas of statistics have long contributed in essential ways to data systems, evaluation, benchmarking, and interdisciplinary research infrastructure, especially in biostatistics, survey methodology, and parts of applied machine learning. At the same time, the level of engagement with large-scale engineering workflows, modern AI pipelines, and deployment-oriented evaluation varies substantially across subfields and application domains. Recognizing this heterogeneity is important both for acknowledging where statistics has already been leading and for identifying areas where broader adaptation could be especially beneficial.

This framing also sharpens what “understanding” should mean in statistics. The goal is not abstraction for its own sake, but actionable understanding grounded in predictive performance, causal insight, and decision relevance. Rigor, elegance, and mathematical structure matter, but in some parts of the field they can become counterproductive when they incentivize narrow assumptions that rarely hold in practice or reward proofs confined to overly sanitized settings. The challenge, then, is to preserve our core identity—principled learning, uncertainty quantification, and deep engagement with domain science—while expanding our willingness to adapt methods, operate within messier data and engineering workflows, and judge success by utility in solving consequential problems.

Large language models (LLMs) (Vaswani et al. 2017; Chang et al. 2024; Song, Segal, and Xing 2024; Deng et al. 2025; Liu et al. 2025) crystallize why this shift is urgent. They are trained on massive, heterogeneous data with minimal distributional assumptions, optimized via generic self-supervision (next-token prediction), and delivered through compute-efficient architectures (transformers) plus substantial engineering layers (fine-tuning, retrieval, alignment). By classical standards, this ecosystem, with trillions of parameters, low-precision computation, extensive data filtering, and post hoc modification, operates far outside the tidy settings in which many theoretical guarantees are typically derived. Yet LLMs achieve striking empirical performance across tasks and domains, signaling that similarly engineered, large-scale empirical models may increasingly reshape areas where statistics has historically been central and where statisticians therefore have strong reasons to engage more directly. Precisely for that reason, statisticians should engage with these systems directly, expressing our comparative advantage, mathematical thinking, careful evaluation, uncertainty reasoning, and sensitivity to data quality and bias, in forms that match modern AI practice, not only in traditional publication formats.

A constructive way to organize this agenda is a system-level view that links, end to end, three pillars: (i) diverse and fair data, (ii) robust statistical/machine learning (ML)/AI methods together with the infrastructure that makes them runnable, scalable and feasible to implement, and (iii) interpretable, actionable

outputs. To make this system-level view concrete, consider a modern healthcare prediction pipeline based on electronic health records. Statistical thinking enters at multiple stages: defining clinically meaningful cohorts and outcomes during data curation; addressing missingness, measurement error, bias, and confounding during preprocessing; developing predictive models with appropriate regularization and uncertainty quantification; evaluating transportability and performance under distribution shift; and monitoring model behavior after deployment. This example illustrates that statistical contributions span the full lifecycle of data-driven systems rather than a single modeling step (Markowetz 2024; Van Calster et al. 2026). A key move is to complement isolated methodological advances with solutions that can also be executed repeatedly, securely, and reliably at scale. This system-level mindset begins with “thinking big” about the operating regime of modern empirical modeling, multi-task problems, data measured in billions or trillions, and computation spanning large GPU clusters, which forces explicit tradeoffs among explainability, actionability, and operational performance, and encourages a more experience-grounded notion of knowledge in which reliable protocols and workflows can be as meaningful as compact equations.

This perspective also discourages over-fixation on abstract labels such as “intelligence,” “optimality,” “convergence,” or even “causality” when these terms are used as slogans and risk substituting terminology for diagnosis. The more productive stance is to interrogate how systems behave on real data, under real constraints, and within real decision pipelines. That implies focusing on original, messy problems rather than forcing premature abstraction, and treating the “last mile” of production, deployment, monitoring, and real-world action as part of the scientific contribution rather than an engineering footnote. It also requires stepping outside disciplinary echo chambers by engaging with data generators, engineers, and end users to understand how data are produced and transformed, how models are operated, and how outputs are consumed and acted upon.

These principles translate into practical priorities under each pillar. On the data pillar, fairness must be a first-order design objective: in many big-data settings, bias dominates variance, and simply increasing sample size does not repair non-representativeness but amplifies inferential errors. This is evident in the portability gaps of polygenic risk scores trained largely on European-ancestry GWAS, underscoring the need for both data diversification and algorithmic remedies such as transfer learning. On the methods-and-infrastructure pillar, robustness is inseparable from usability: automation across data access, pipeline construction, evaluation, and deployment is essential to ensure analyses can be executed repeatedly, auditably, and securely. As biomedical resources shift toward cloud-native “bring researchers to the data” ecosystems, statisticians must develop practical fluency with cloud platforms, workflow languages, reproducible pipelines, and packaging and deployment practices so our methods can run within secure, workspace-based environments. On the outputs pillar, the goal is not just predictive accuracy but results that domain experts can interpret, verify, and trust. In many cases, inference is the main objective, and prediction is simply a tool to support it—for example, prediction-powered inference uses machine-learning predictions to improve estimation and hypothesis testing.

Technically, this shift also implies broadening the statistical toolkit toward the paradigms and mathematics that increasingly power modern AI: multi-task learning, federated learning, and reinforcement learning; objectives such as contrastive and reward-based losses; and architectures that implement complex procedures rather than relying primarily on strong priors. Progress is often driven less by distribution fitting and likelihood optimization and more by ideas from differential geometry, information theory, linear algebra, optimization, and variational calculus, complemented by visualization tools that help build intuition (e.g., spectral embeddings and phylogenetic-style representations). These themes point naturally to curriculum updates that reflect how modern systems are actually built and validated.

In sum, the field should judge rigor by dependable performance in real operating conditions, not by abstraction alone. Trust can be earned through methods and workflows that work reliably and repeatedly, whether or not they are accompanied by elegant proofs. The north star is the problem being solved, and whether the full system from data to deployment delivers consistent, interpretable, and actionable value.

3. Data Curation and “Data Work”

Data curation, wrangling, and annotation are not auxiliary “support tasks”; they are foundational scientific work that shapes bias, uncertainty, interpretability, and ultimately the trustworthiness of AI systems (US.NIH.AI 2025; US.NSF.Inf 2025; American Statistical Association 2025). A recurring theme was that statistics will remain influential only if statisticians engage directly with how data are generated, collected, transformed, and governed at scale and if departments realign training, infrastructure, and incentives so that high-quality data resources and reproducible pipelines are valued alongside new methods.

A key cultural obstacle is the persistent division between “model work” and “data work,” reinforced by reward systems that often celebrate new methodology while treating annotation, harmonization, and cleaning as invisible labor. Yet modern AI products, from autonomous systems to large language models, depend critically on these upstream decisions. The discussion highlighted the need to broaden what counts as a major statistical contribution by building shared data standards and infrastructures, enabling cross-institutional comparability, and creating durable community resources that compound in value over time, for example, large-scale electronic health record (EHR) ecosystems and common data models. Closely related is the call for statisticians to lead, not just analyze, large data initiatives, since defining cohorts, outcomes, and measurement processes is inseparable from valid inference and responsible deployment.

Two examples illustrated why ambitious data work can be field-shaping. Large, carefully annotated benchmarks helped catalyze breakthroughs by making previously underpowered methods viable and comparable across groups, showing that “dataset building” can be a research contribution on par with algorithmic innovation. In biomedicine, the rapid expansion of whole-genome sequencing and biobanks has created an analogous need for large-scale functional annotation and user-facing tools that make such annotations searchable, reusable, and cheap

enough to support routine downstream analysis. The broader point was that the “last mile” matters: annotation is only transformative when it is integrated into analysis workflows, engineered for reuse, and delivered with interfaces and indexing that real users can operate.

This emphasis on data work also reframes both methodology and training. On the technical side, trustworthy learning requires bringing statistical thinking into the full data lifecycle: uncertainty arising from measurement and acquisition, sensitivity to data bias and missingness, and systematic evaluation and monitoring once models are deployed. On the practical side, large modern datasets are high-dimensional, multi-modal, and heavily preprocessed (tokenization, alignment, embeddings), often cloud-hosted, and increasingly augmented with synthetic data—conditions that demand stronger “intermediate” computational literacy (e.g., APIs, version control, reproducible workflows) and closer collaboration with data engineers and domain teams. The panel also stressed that scalable, cloud-native pipelines are now part of scientific validity: if analyses cannot be executed repeatedly, securely, and cost-effectively, then even elegant methods will struggle to make real impact.

Finally, the conversation pushed for a more pragmatic, data-centric posture toward credibility. Rather than treating rigor as synonymous with formal proofs in idealized settings, the panel argued for rigor as dependable performance under real operating conditions—validated through careful data construction, transparent pipelines, and reproducible results. In this view, building trustworthy systems means stepping out of disciplinary echo chambers, staying close to real data-generation processes, and investing in the data and infrastructure work that makes end-to-end science possible.

4. Engaging with Innovations in Empirical Modelling

Statisticians should engage directly with modern machine learning and large-scale empirical modeling (Breiman 2001; Lin et al. 2025; Cai 2025; He et al. 2025), rather than treating these developments as external or “purely engineering.” A central message of the town hall was that the statistical lens, such as uncertainty quantification, bias and causal thinking, interpretability, and rigorous evaluation, remains indispensable for turning high-performing models into systems that are robust, generalizable, and scientifically useful, especially in high-stakes settings. But contributing at that level requires meeting modern ML on its own terrain: understanding how models are trained and engineered at scale, and collaborating closely with computer scientists and data engineers so statistical ideas can be embedded into real pipelines rather than appended after the fact.

Several panelists underscored that this engagement demands an openness to new regimes where classical intuitions can mislead. One illustration was the initial “overfitting” reaction many statisticians had to LLMs with parameter counts that dwarf traditional sample-size notions, followed by the realization that contemporary practice often proceeds by building systems that work and only later developing the theory that explains when and why. Phenomena such as double descent further challenge the one-directional story of capacity and generalization, suggesting that some familiar graduate-school heuristics need to

be revisited in the context of modern optimization, scale, and training dynamics. This is less an argument to abandon theory than a call to re-sequence and reframe it: theory that stays connected to actual operating regimes, and inference targets that remain meaningful when models are trained on heterogeneous, internet-scale corpora.

At the same time, the discussion emphasized that engagement should not get trapped in abstract debates about labels (“intelligence,” “optimality,” “guarantees”) divorced from measurable utility. A pragmatic stance within today’s ML community is to define progress operationally, by performance on well-specified tasks and by behavior in deployment, while using statistics and applied mathematics as enabling tools (e.g., sampling, search, approximate inference, and monitoring). From this viewpoint, what matters most is not whether a system conforms to a tidy probabilistic story, but whether it performs reliably under real constraints, can be stress-tested, and can be improved systematically. Newer ideas like “reasoning via simulation” and world-model-based planning sharpen this point: they may rely on sophisticated statistical machinery, but their value is ultimately judged by downstream decisions, robustness, and repeatable gains rather than philosophical purity.

Taken together, the panel’s message was that statisticians can make distinctive contributions by (i) clarifying what should be estimated and evaluated in these systems, including uncertainty and failure modes, (ii) diagnosing and mitigating bias and confounding in training data and deployment settings, and (iii) strengthening interpretability and auditing so models can be trusted in consequential decision pipelines. The fastest path is not to stand outside and critique, but to participate, developing fluency with contemporary ML workflows and helping shape the next generation of empirical modeling so it is not only powerful, but also principled and dependable in the real world.

5. Next-Generation Talent in Large-Scale Applications

Training the next generation of statisticians must shift from a course-centric model to an end-to-end, systems-oriented apprenticeship for data-and-AI practice. Beyond classical foundations, trainees need durable skills in data curation and annotation, modern ML, scalable computing, and principled inference about uncertainty, bias, and interpretability. The panel emphasized that these capabilities are best developed through interdisciplinary curricula and experiential structures—team-based capstones on real (often messy) datasets, cross-mentorship with computer science and domain experts, and reproducible software practices that make methods implementable, shareable, and auditable (Lin et al. 2025; He et al. 2025; Scharfmann, Marx, and Fleming 2025). Equally important, departments must provide the surrounding ecosystem—shared infrastructure, staff-scientist support, and clear career pathways, often in partnership with industry, so trainees learn not only to propose ideas, but to deliver trustworthy systems that run reliably at scale.

A recurring recommendation was to rebalance education around the “three C’s”: communication, collaboration, and computation. That means strengthening students’ ability to work in teams, translate statistical reasoning to diverse stakeholders, and operate within modern engineering workflows. It also requires

updating what counts as core theory: rather than relegating the mathematical underpinnings of AI or stochastic thinking about graphs and networks to special-topics electives, the panel suggested making them expected components of graduate training, alongside at least a baseline introduction to deep learning concepts such as transformers and autoencoders. Early pipeline programs (e.g., summer institutes and REUs) were highlighted as strategic levers for attracting quantitatively strong students who might otherwise default to computer science, and for bringing them into statistics and biostatistics before disciplinary identities harden.

At the same time, the panel cautioned against “tool overload.” The goal is not to stuff curricula with every fashionable method, but to train curiosity and problem-solving by immersing students in real problems and letting tool choice follow the demands of the task. Importantly, the panel also emphasized that the use of AI should not be treated punitively; instead, students should learn to use AI models responsibly and effectively so that routine tasks are delegated while human understanding, judgment, and accountability remain central. This implies treating curriculum design as an evolving engineering process: regularly auditing what is taught against what graduates actually use, being willing to remove legacy material before adding new content, and drawing inspiration from how rapidly AI and CS programs continually reinvent themselves. One illustrative vision is an AI-native program that combines deep technical training (e.g., ML, NLP, CV, robotics) with cross-disciplinary preparation in design, communication, and even business and entrepreneurship. The panel suggested that statistics programs should emulate this spirit by expanding breadth while staying anchored in statistical principles.

Because PhD timelines are unlikely to lengthen, the panel argued for a pragmatic strategy: shrink the “intersection” (what every statistician must know) so the “union” (the community’s collective capabilities) can grow. In practice, this means preserving a shared base in statistical reasoning, study design, and uncertainty, while encouraging diverse, complementary skill profiles rather than cloning a single archetype. As a minimum standard, “intermediate” computational literacy (version control such as Git, reproducible workflows, basic software engineering habits, and APIs for data access) should become routine so new methods can be implemented, disseminated, and adopted. Finally, training should cultivate system-level thinking: moving beyond one-off case studies to understanding how methods behave when deployed repeatedly in pipelines, platforms, and decision systems, potentially including LLM-enabled tools as part of modern statistical workflows.

6. Engaging AI Stakeholders

Statistics should engage AI stakeholders more proactively by translating its core strengths, including uncertainty quantification, bias and causal reasoning, interpretability, and reproducible evaluation, into the language of real systems, measurable impact, and operational risk. The panel argued that influence will come less from episodic “consulting” and more from building a durable ecosystem: sustained partnerships across academia, industry, government, and national labs; shared data

resources and benchmarking infrastructure; and training pathways that let people move fluidly between academic research and deployed AI practice (The White House 2025). Just as important, the field must advocate for funding, review, and promotion structures that recognize end-to-end contributions, so statistical principles shape not only methods papers, but also the data, software, and evaluation pipelines that determine whether AI is trustworthy in the real world.

A recurring theme was that statisticians should seek institutional leadership roles that provide a “big-picture” seat at the table where AI investments are made. Participating in university-wide data strategy, AI, and high-performance computing initiatives creates visibility with senior administrators and opens access to resources that far exceed what a single department can typically marshal. The same logic applies nationally: partnerships with supercomputing-centered national laboratories can anchor statistics in the compute and infrastructure decisions that increasingly determine what research is feasible. Within these collaborations, the panel emphasized that our comparative advantage is clearest when we focus on system-critical questions such as how training-data bias affects transportability to deployment settings, how sampling and experimental design principles can be adapted for AI, how to quantify uncertainty and monitor drift, and how to evaluate and compare models rigorously. They also urged realism and humility: statisticians may not be the ones building the next frontier LLM, and many problems will resist clean theory, but that is precisely why collaboration matters, along with recruiting more engineering-oriented faculty and staff who can help operationalize statistical ideas.

The panel also stressed that credibility with stakeholders ultimately comes from being present from start to finish: learning the real problem context, working closely with the people operating the system, contributing to solution development, and then telling those stories in ways that stakeholders recognize as valuable. This includes adopting the funder’s and decision maker’s perspective by explaining why a major investment should go to a statistical effort, what returns it produces, and what risks it reduces. The message was pragmatic: impact earns autonomy. If we want the freedom to pursue ambitious agendas, we must demonstrate utility through tangible deliverables such as datasets, pipelines, benchmarks, deployed tools, and decision-relevant theory (for example, uncertainty for specific models, learning curves, and operational sample and compute complexity) that speaks directly to how modern AI systems behave in practice.

7. Concluding Remarks

The closing remarks emphasized how valuable it is for statisticians to hear perspectives from outside the field, especially when they name, bluntly, the ways our community can be in denial about what the AI era demands. Several points were described as “solid gold” precisely because they articulated, statement after statement, what statisticians ought to feel “in their bones” but often do not, underscoring a sense of urgency rather than mere intellectual interest.

The speakers then widened the lens to a cultural backdrop, arguing that the media environment itself is reshaping human

attention and cognition. A vivid Olympics scene, an iconic figure surrounded by younger spectators absorbed in their phones, was used to suggest that this level of device-mediated immersion is new and deeply disruptive, consuming attention in ways we do not yet fully understand. These are not simply the same students with new tools; the habits of sustained engagement that shaped earlier generations may be eroding.

From this vantage point came a caution against assuming that “statistics and data science” can be fused by rebranding alone (American Statistical Association 2025; Scharfmann, Marx, and Fleming 2025). The gap is not only technical but also cultural and epistemic: in a world dominated by smartphones, many people may have little capacity or incentive for extended engagement with mathematically intensive, print-era modes of thinking (Lin et al. 2025; Breiman 2001; Cai 2025; He et al. 2025). This connects to the “Gutenberg parenthesis” idea: if mass book culture is giving way to always-on information streams, then mathematics, and statistics as a mathematically grounded discipline, sits inside that parenthesis, raising uncomfortable questions about how knowledge will be learned and valued.

Returning to the five guiding questions of this article, we conclude that the future of statistics in the age of AI will depend on how well the field can respond across all five fronts: evolving its culture toward broader, system-level thinking; recognizing data curation and stewardship as core scientific contributions; engaging more deeply with modern empirical modeling; rethinking training for large-scale, AI-enabled applications; and building sustained partnerships with key AI stakeholders. Together, these themes point to a common challenge: extending the field’s traditional strengths into a rapidly changing, data-centric environment. These responsibilities are not entirely new, but the scale and speed of AI make them newly urgent. If statistics can meet this moment with both rigor and adaptability, it will not merely remain relevant, but help shape the data-centric future.

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The authors report there are no competing interests to declare.

ORCID

Jian Kang  <https://orcid.org/0000-0002-5643-2668>

Rebecca Nugent  <https://orcid.org/0000-0001-5503-7394>

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